

Vishwash Agarwal

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Professional Summary

C/C++ and Python developer with strong foundations in Data Structures, Operating Systems, and Computer Architecture. Experienced in debugging, testing, and improving performance of real-time and ML-based systems.

Education

Vellore Institute of Technology, India

Integrated MTech in Data Science

St. Joseph's School, Gorakhpur

Class 12 – 91.6% — Class 10 – 93.0%

2022–Present

CGPA: 8.64

2020–2022

Skills

Languages: C, C++, Python, JavaScript

Core CS: Data Structures, Algorithms, Operating Systems, Computer Architecture, DBMS, OOP

Systems: Debugging, Testing, Performance Tuning, APIs, Linux, Git

Tools: TensorFlow, OpenCV, NumPy, Pandas, Firebase, AWS

Experience

Machine Learning Intern — Prodigy InfoTech

Nov 2025 – Dec 2025

- Implemented MobileNetV2 inference pipeline achieving 91.75% validation accuracy on custom dataset.
- Built authentication, Firestore logging, and REST APIs enabling reliable multi-user sessions.
- Reduced latency in OpenCV + TensorFlow vision pipeline through preprocessing and memory optimizations.
- Conducted experiments, hyperparameter tuning, and systematic debugging to improve stability and throughput.
- Performed testing, benchmarking, and deployment validation across modules before production release.

Projects

Indian Food Recognition & Calorie Tracker

[Live] — [GitHub]

- Designed MobileNetV2-based image classifier achieving 91.75% validation accuracy on custom dataset.
- Built preprocessing, batching, and inference pipeline improving throughput and memory efficiency.
- Integrated Firebase authentication, Firestore database, and REST APIs for scalable multi-user tracking.
- Logged user activity and calorie data with automated validation and error handling for reliability.

Hand Gesture Recognition System

[Live] — [GitHub]

- Engineered real-time CNN-based gesture recognition using OpenCV webcam streams.
- Implemented normalization and augmentation to address domain shift and lighting variability.
- Reduced inference latency enabling smooth real-time interaction and higher FPS.
- Structured modular pipeline for easy retraining, testing, and deployment of updated models.

YouTube Video Summarizer (LLM Pipeline)

[GitHub]

- Developed transcript extraction → chunking → summarization pipeline using Gemini API.
- Implemented caching, batching, and retry logic to minimize API calls and failures.
- Improved response time by 40–60% through efficient token management and parallel processing.
- Exported outputs to structured formats (.txt/.json) and enabled interactive chat with video content.

Problem Solving

Solved 120+ LeetCode problems across arrays, trees, graphs, and dynamic programming.

Achievements

- President, Basketball Club – Led 25-member team and organized inter-school tournaments.
- Gold Medallist – Science, Cyber, and NASA Olympiads for academic excellence and problem-solving skills.
- Vice-Captain, Green House – Coordinated events and improved team participation.
- Runner-Up – Speech Competition demonstrating communication and public speaking ability.